Project Design Phase-II
Data Flow Diagram & User Stories

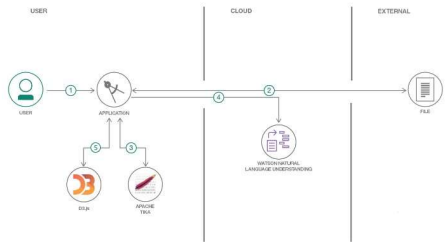
Date	31 January 2025
Team ID	
Project Name	PlantCare AI: Advanced Plant Disease Detection Using Transfer Learning
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

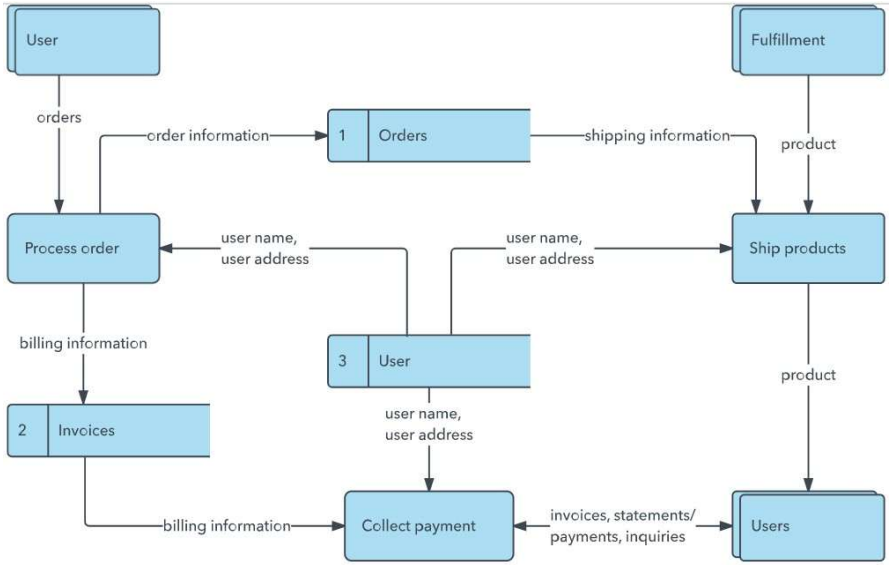
Example: (Simplified)

Flow

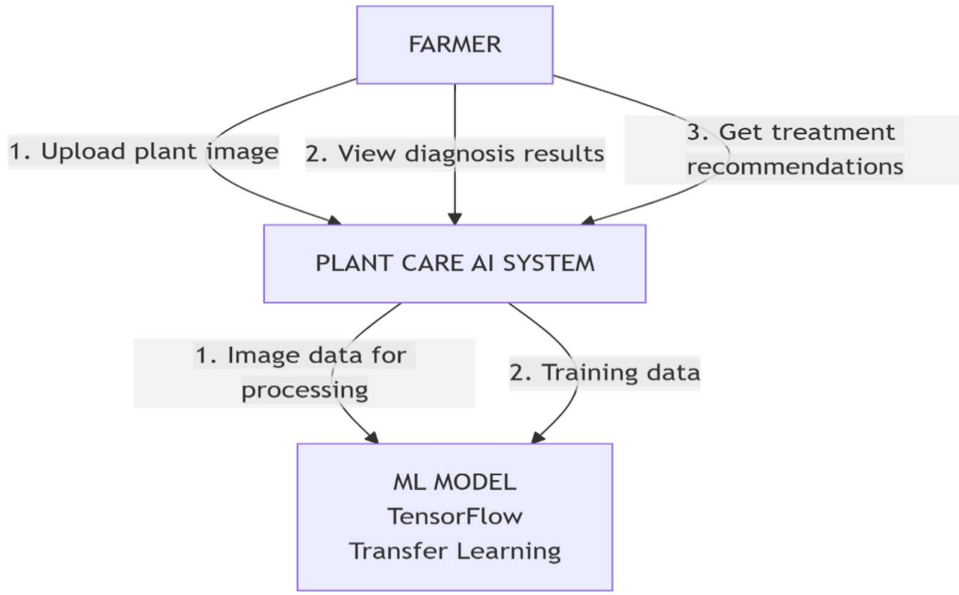


1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

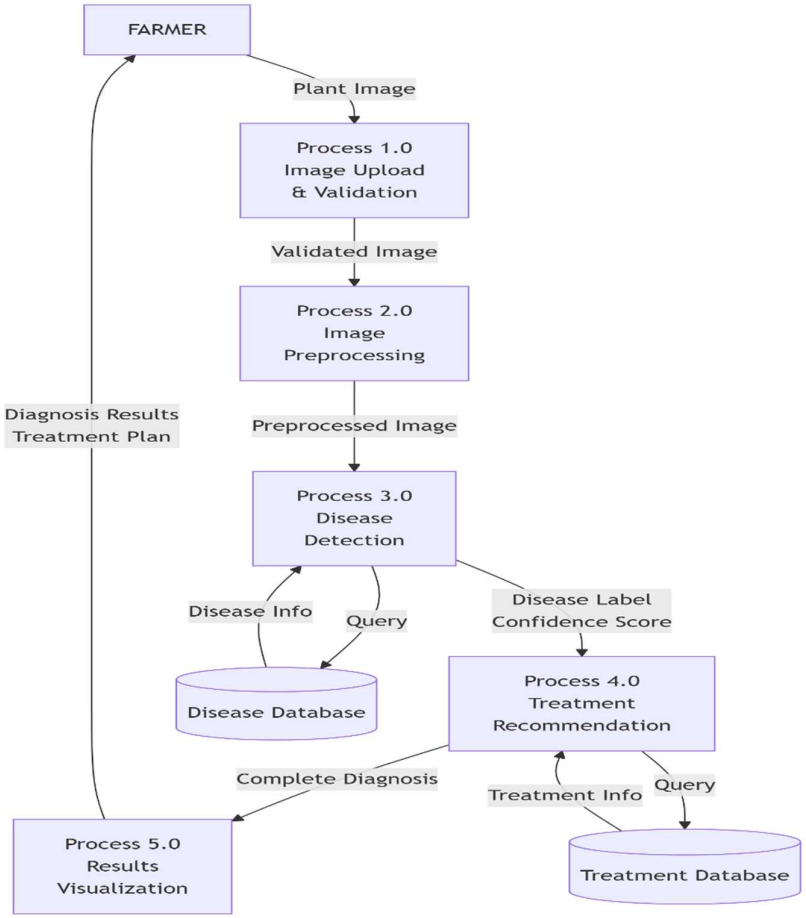
Example: DFD Level 0 (Industry Standard)



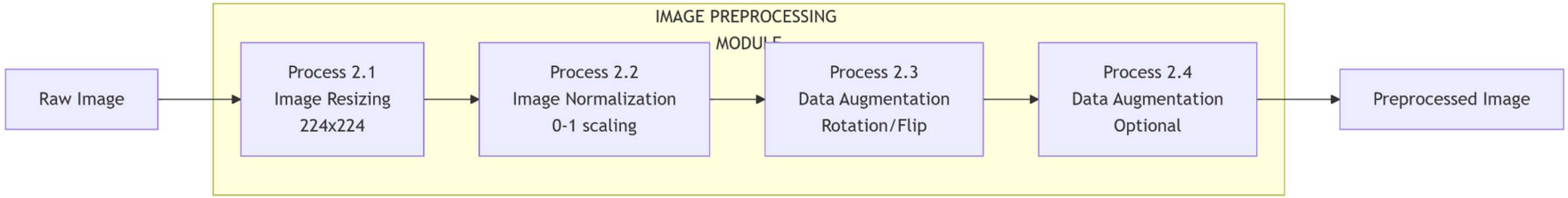
1. Context Diagram (Level 0)



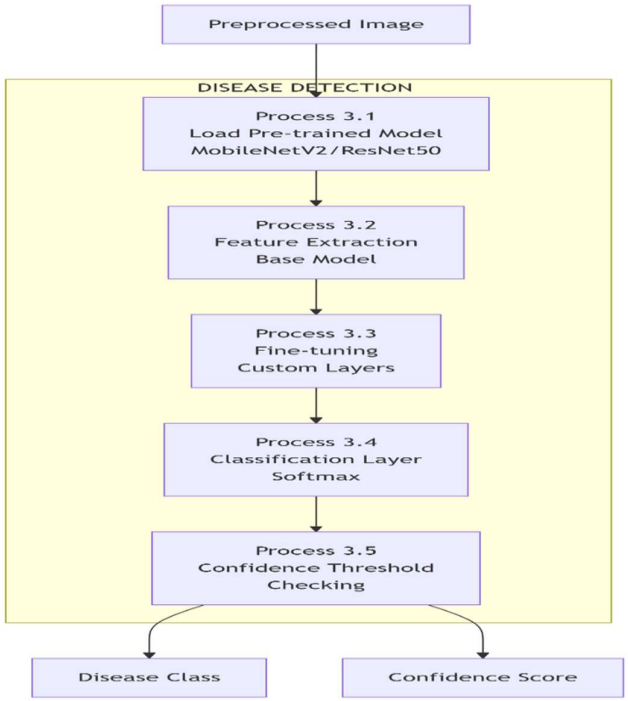
2. Level 1 DFD - Main Processes



3. Level 2 DFD - Image Preprocessing (Expanded)



4. Level 2 DFD - Disease Detection (Expanded)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Mobile user)	User Registration and Profile Management	USN-1	As a new user, I want to register an account so that I can save my plant diagnosis history	• User can sign up with email/password • Google/Facebook OAuth option available • Email verification sent • Profile created successfully	High	Sprint-1
		USN-2	As a registered user, I want to log in securely so that I can access my account	• Login with email/password • "Remember me" option • Password reset functionality • 2FA option for security	High	Sprint-1
		USN-3	As a user, I want to update my profile information so that my details remain current	• Edit name, email, password • Upload profile picture • Save changes confirmation	Medium	Sprint-2
		USN-4	As a user, I want to delete my account so that my data is removed from the system	• Account deletion option • Confirmation dialog • Data anonymization • Final email confirmation	Low	Sprint-1
	Image Capture and Processing	USN-1	As a user, I want to capture a photo of my plant using the app camera so that I can get disease diagnosis	• Camera opens within app • Auto-focus and flash options • Capture button works • Preview before processing	High	Sprint-1

		USN-2	As a user, I want to upload existing photos from my gallery so that I can diagnose previously taken pictures	• Gallery access permission • Image selection interface • Multiple image selection • Preview selected images	High	
		USN-3	As a user, I want to see a preview of my captured image before processing so that I can retake if needed	• Full-screen preview • Retake option • Confirm button • Image quality indicator	Medium	
		USN-4	As a user, I want to crop my image to focus on the affected leaf so that diagnosis is more accurate	• Crop tool interface • Aspect ratio lock • Reset option • Preview cropped image	Medium	
	Disease Detection	USN-1	As a user, I want to get instant disease detection results so that I can take quick action	• Loading indicator during processing • Results within 5 seconds • Clear disease name display • Confidence percentage shown	High	
		USN-2	As a user, I want to know the confidence level of the diagnosis so that I can trust the results	• Percentage display • Color-coded confidence (green/yellow/red) • Explanation of confidence meaning • Option for second opinion	High	
		USN-3	As a user, I want to see similar images of the detected disease for comparison	• Gallery of similar cases • Side-by-side comparison • Zoom functionality • Source attribution	Medium	
		USN-4	As a user, I want the app to work offline so that I can diagnose plants in areas with no internet	• Offline mode indicator • Basic model available offline • Queue for cloud sync • Notification when online	High	

	Treatment Recommendations	USN-1	As a user, I want to get treatment recommendations for the detected disease so that I can help my plant recover	• Treatment steps clearly listed • Organic options first • Chemical alternatives • Safety precautions	High	
		USN-2	As a user, I want to see both organic and chemical treatment options so that I can choose based on preference	• Tabbed interface (Organic/Chemical) • Pros and cons for each • Cost estimates • Availability information	Medium	
		USN-3	As a user, I want to get preventive measures so that I can avoid future infections	• Preventive tips list • Seasonal advice • Watering recommendations • Fertilizer schedule	Medium	
		USN-4	As a user, I want treatment recommendations based on my local weather so that they are relevant to my area	• Weather integration toggle • Location permission • Localized advice • Weather-based warnings	Medium	
	History and Tracking	USN-1	As a user, I want to view my diagnosis history so that I can track my plants' health over time	• Chronological list view • Thumbnail images • Disease names displayed • Date stamps	High	
		USN-2	As a user, I want to search my history by plant type or disease so that I can find specific cases	• Search bar • Filter by date • Filter by disease • Filter by plant	Medium	
		USN-3	As a user, I want to add multiple plants to my profile so that I can manage my garden	• Add plant form • Plant name and type • Photo upload • Location tagging	Medium	

		USN-4	As a user, I want to set reminders for treatment follow-up so that I don't miss important steps	• Reminder creation • Date/time picker • Notification delivery • Snooze option	Low	
	Social and Community Features	USN-1	As a user, I want to share my diagnosis results with others so that I can get second opinions	• Share button • Share via WhatsApp, Email, SMS • Include images and results • Privacy options	Medium	
		USN-2	As a user, I want to see community posts about plant diseases so that I can learn from others	• Community feed • Filter by disease/plant • Like and comment • Report inappropriate content	Low	
		USN-3	As a user, I want to contribute my diagnosed images to improve the AI model so that the app gets better	• Opt-in checkbox • Anonymous contribution • Thank you message • Impact metrics	Medium	
		USN-4	As an expert, I want to validate user diagnoses so that the community gets accurate information	• Expert verification badge • Correction tool • Explanation field • User notification	Low	
Administrator	Admin and Expert Features	USN-1	As an admin, I want to view system analytics so that I can monitor usage and performance	• Dashboard with metrics • User growth chart • Diagnosis count • Error rate monitoring	High	
		USN-2	As an admin, I want to manage the treatment database so that information stays current	• CRUD operations for treatments • Version control • Review before publish • Effective dates	High	
		USN-3	As an expert, I want to update disease information so that users get accurate data	• Edit disease profiles • Add new diseases	Medium	

				• Upload reference images • Cite sources		
		USN-4	As an admin, I want to manage user reports and feedback so that the community remains healthy	• Report queue • Action options (warn/suspend) • Appeal process • Audit log	Medium	
	Model Improvement and Feedback	USN-1	As a user, I want to provide feedback on diagnosis accuracy so that the model can improve	• Feedback buttons (correct/incorrect) • Comment field • Optional image upload • Thank you note	Medium	
		USN-2	As a data scientist, I want to collect user-validated images so that I can retrain the model	• Feedback database • Export functionality • Quality filters • Annotation tools	High	
		USN-3	As a data scientist, I want to A/B test model versions so that I can deploy the best performing model	• Model versioning • User segmentation • Performance metrics • Rollback capability	High	
		USN-4	As a system, I want to automatically retrain models periodically so that accuracy improves over time	• Scheduled retraining • New data integration • Validation pipeline • Deployment automation	Medium	